

```
#include <Wire.h>

int offset =20;

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

int A,B,C,D;

int a=0;

int b=0;

int c=0;

float h;

float t;

#define gas 9

#define buzzer 8

#define relay 10


#include <SoftwareSerial.h>

void setup()
{
    Serial.begin(9600);

    lcd.init();

    lcd.backlight();

    lcd.setCursor(0,0);
    lcd.print("ALCOHOLE SENSING ALERT ");
    lcd.setCursor(0,1);
    lcd.print("SYSTEM ENGINE");
    delay(2000);
    lcd.clear();


    pinMode(buzzer,OUTPUT);

    pinMode(gas,INPUT);

    pinMode(relay,OUTPUT);

    digitalWrite(relay,0);
```

```
}
```

```
void loop()
```

```
{
```

```
    lcd.clear();
```

```
    lcd.setCursor(0,0);
```

```
    lcd.print("ALCOHOLE-: ");
```

```
    lcd.setCursor(7,0);
```

```
    lcd.print(voltageValue());
```

```
    lcd.setCursor(11,0);
```

```
    lcd.print("ppm");
```

```
    delay(2000);
```

```
}
```

```
String voltageValue(){
```

```
    A=digitalRead(9);
```

```
    a=map(A,1,0,0,78);
```

```
    if (A==0)
```

```
    {
```

```
        lcd.clear();
```

```
    lcd.setCursor(0,0);
```

```
lcd.print("ALCOHOLE DETECTED");
```

```
digitalWrite(relay,0);
```

```
digitalWrite(buzzer,1);
```

```
delay(1000);
```

```
digitalWrite(buzzer,0);
```

```
lcd.setCursor(0,1);
```

```
lcd.print("STOPPED");
```

```
delay(2000);
```

```
lcd.clear();
```

```
}
```

```
else{
```

```
    digitalWrite(relay,1);
```

```
}
```

```
return String(a);
```

```
}
```